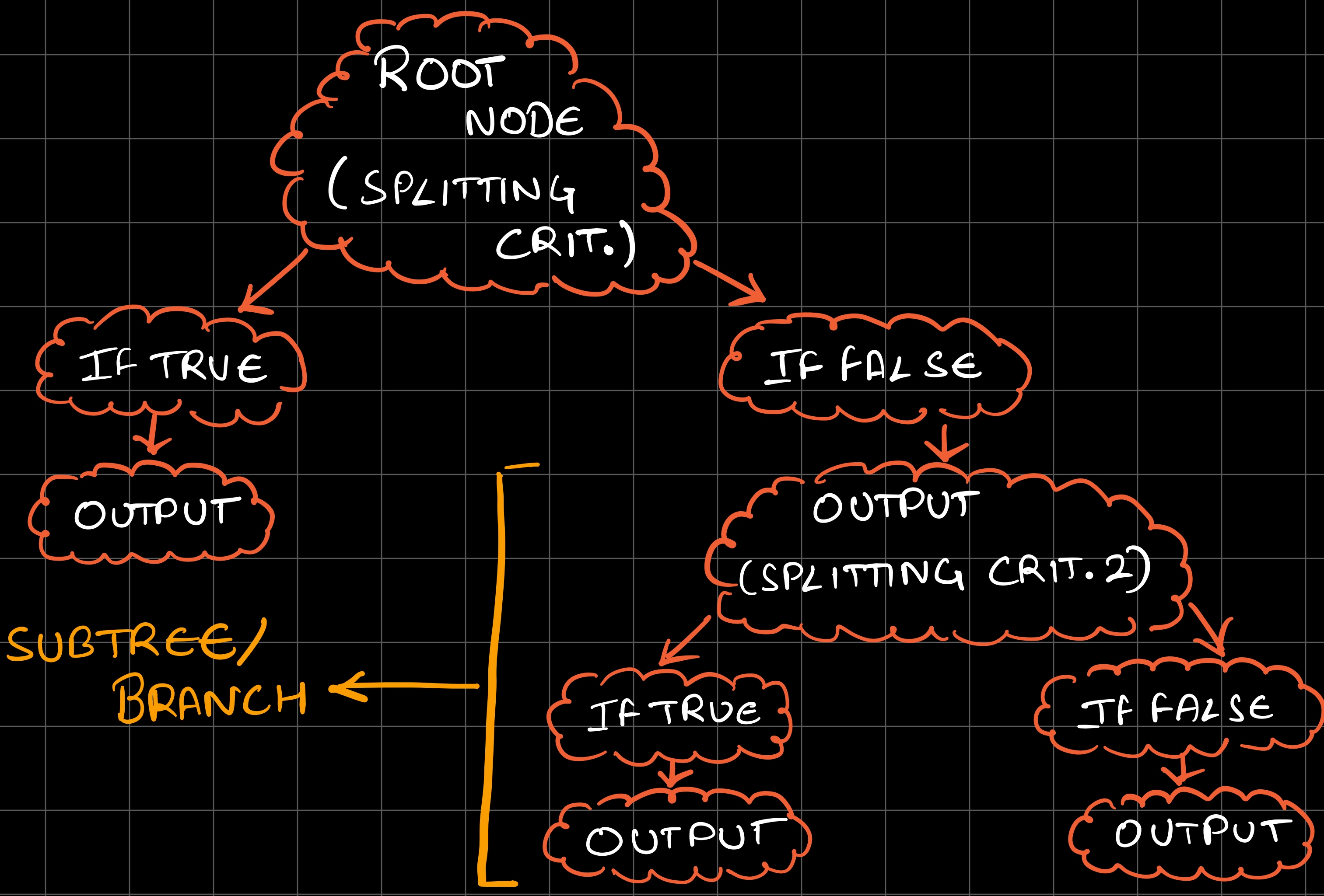
DECISION TREE

Its like a large if-else ladder which resembles a tree.



ADVANTAGES :

⇒ Intutive and easy to use.

⇒ Minimal data preperation.

DISADVANTAGES:

⇒ Overfitting

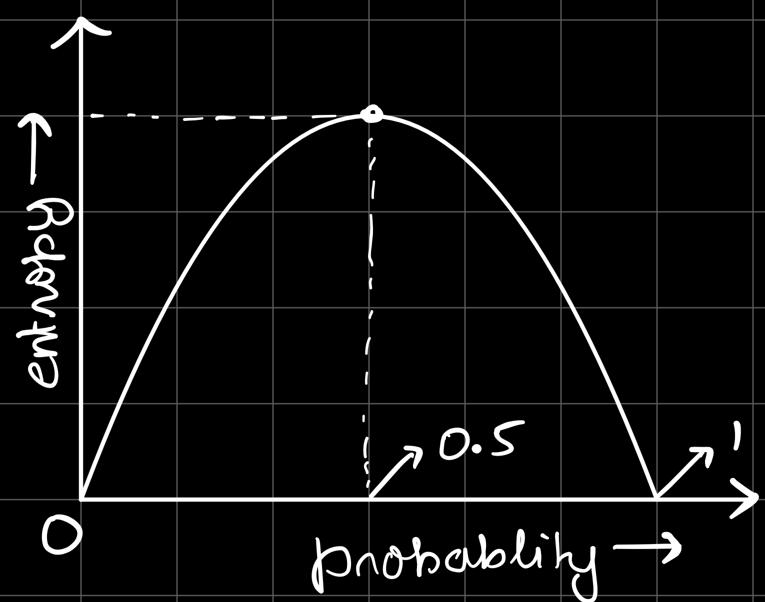
⇒ Prone to errors in imbalanced dataset.

ENTROPY: Degree of randomness

$$\boxed{\epsilon = \sum -p_i \log_2 p_i}$$

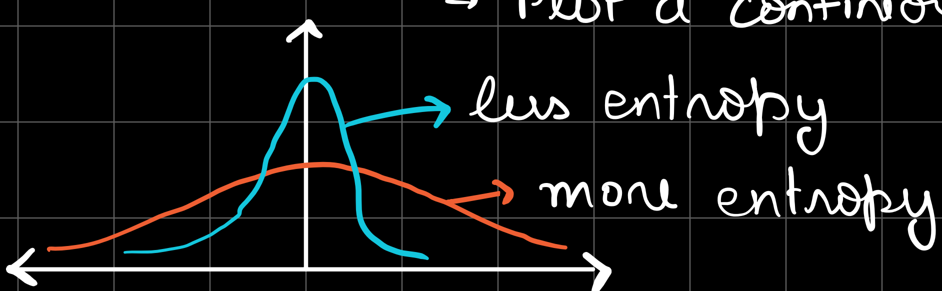
⇒ p_i is the frequentist probablity

⇒ ↑ Entropy, ↑ Uncertainity



FOR CONT. DATA:

⇒ Plot a continious histogram.



⇒ KDE Plot.

GAIN : $\boxed{\text{GAIN} = E(\text{PARENT}) - \sum(\text{WEIGHT AVG})\, E(\text{CHILD})}$

⇒ Find column with more gain and then that data is used to do the splitting.

GINI IMPURITY :

$$\boxed{G_i = 1 - \left(\sum P_i^{2}\right)}$$

⇒ GINI is computationally harder as calculating log is more comp. than squares.